

Summary of *General Agents Contain World Models*

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The follow-up to *Robust Agents Learn Causal World Models*: that paper asked what an agent must know to stay near-optimal when the *environment* shifts; this one asks what it must know to pursue many *goals* in a fixed environment. Again competence forces a model—now predictive rather than causal—with accuracy forced up by the *depth* of the goals achieved.

Basic setup

The environment is a *controlled Markov process* (cMP): an MDP with no reward and no discount, given by a tuple $(\mathcal{S}, \mathcal{A}, P)$ with finite state space $\mathcal{S}$, action space $\mathcal{A}$, and transition function

$$P_{ss'}(a) = P(S_{t+1} = s' \mid S_t = s, A_t = a).$$

Assumption 1. The cMP is finite, stationary (transition probabilities do not change over time), *communicating* (every state is reachable from every other under some policy), has at least two actions, and is fully observed.

A *world model* is an approximation $\hat{P}$ of the transition function with bounded error, $|\hat{P}_{ss'}(a) - P_{ss'}(a)| \leq \varepsilon$ for all s, s', a . Rolled forward, such a one-step predictor predicts the environment under any policy.

An *agent* is a deterministic goal-conditioned policy, a map $\pi(h_t, \psi) = a_t$ from the history h_t and the goal ψ to an action. Two objects must be kept apart: $P_{ss'}(a)$ is the *environment's* transition law and is never an input to the policy. The theorem is striking precisely because P can nonetheless be reconstructed from the policy's behaviour. The paper's question is whether an agent that pursues a rich family of goals must, in this sense, determine a model of P .

Goals

Goals are built in three layers, from linear temporal logic.

Base goals. A base goal is $\varphi = \mathcal{O}([(s, a) \in \mathbf{g}])$, where $\mathbf{g}$ is a set of goal state–action pairs and $\mathcal{O}$ is one of three temporal operators,

$$\top = \text{now}, \quad \bigcirc = \text{next transition}, \quad \diamond = \text{eventually}.$$

These differ in horizon and in *what they constrain*. *Now* and *next* look alike but do different jobs: the atomic predicate is on a state–action pair, so $\top[(s, a)]$ says “be in s and take a at the current step”—it pins down the agent's own action, which it controls—whereas $\bigcirc[(s', \cdot)]$ says “be in s' at the next step”—it pins down the environment's response, which it does not. The two-stage goal $\langle \top[(s, a)], \bigcirc[(s', \cdot)] \rangle$, “do a in s and land in s' ”, is exactly the unit trial the extraction repeats: without *now* a goal could not force a particular action, without *next* it could not refer to the outcome. *Eventually* is different again: “eventually visit a state in $\mathbf{g}$ ” can range over an arbitrarily long trajectory. Long-horizon *structure*, as opposed to long duration, comes from sequential composition.

Sequential goals. $\psi = \langle \varphi_1, \dots, \varphi_n \rangle$ requires the sub-goals to be achieved in order, φ_i before φ_{i+1} . Its *depth* is $\text{depth}(\psi) = n$: the number of sequentially composed base goals. This is *not* the number of environment time steps—a single $\diamond$ sub-goal may take arbitrarily many transitions—but the number of stages the agent must chain.

Composite goals. $\psi = \bigvee_{i=1}^m \psi_i$ is a disjunction of sequential goals; its depth is the largest depth among them. Ψ_n denotes all composite goals of depth at most n .

Write $\tau \models \psi$ when trajectory τ satisfies ψ ; the success probability of a policy from s_0 is $P(\tau \models \psi \mid \pi, s_0)$. Two features of this goal language do all the work later: depth n measures how many stages a task chains, and a disjunction offers the agent *alternatives it must choose between*, which is what makes its preferences observable. These are not arbitrary preferences: choosing between incompatible branches reveals which has the higher *achievable* success probability, which is why behaviour carries numerical information about the dynamics.

Agents and regret

An *optimal* goal-conditioned agent maximises success on every goal from every start state, $\pi^* = \arg \max_{\pi} P(\tau \models \psi \mid \pi, s_0)$ for all $\psi \in \Psi$ and all s_0 with $P(s_0) > 0$.

The object of study is weaker, and it is the paper’s organising definition. A *bounded goal-conditioned agent* with maximum goal depth n and failure rate $\delta \in [0, 1]$ is any policy satisfying

$$P(\tau \models \psi \mid \pi, s_0) \geq (1 - \delta) \max_{\pi'} P(\tau \models \psi \mid \pi', s_0) \quad \text{for all } \psi \in \Psi_n.$$

In words: *competence* is how close the agent’s probability of satisfying each goal comes to the best achievable, uniformly over every goal in Ψ_n . This is an external performance criterion—the agent need not know the optimum, or that one exists—and it is the only assumption on the agent. It is the counterpart of δ -optimality across domain shifts, with goals playing the role interventions played; the regret is *multiplicative* (a fraction of the optimum) rather than additive.

Main theorems

Theorem 1 (world models from bounded regret). Let the environment satisfy Assumption 1 and let π be a bounded goal-conditioned agent with failure rate δ on all goals in Ψ_n , with $n > 1$. Then π fully determines a model $\hat{P}$ with

$$|\hat{P}_{ss'}(a) - P_{ss'}(a)| \leq \sqrt{\frac{2P_{ss'}(a)(1 - P_{ss'}(a))}{(n-1)(1-\delta)}} \sim O\left(\frac{\delta}{\sqrt{n}}\right) + O\left(\frac{1}{n}\right) \quad (\delta \ll 1, n \gg 1).$$

bounded regret at depth $n \implies$ policy determines a world model to $O(\delta/\sqrt{n}) + O(1/n)$.

Three readings. The error vanishes as the agent improves ($\delta \rightarrow 0$) *or* as its goals deepen ($n \rightarrow \infty$); any fixed $\delta < 1$ is overcome by large enough n , so the policy of even a mediocre agent that chains long tasks determines an increasingly accurate model. The bound is absolute, so rare transitions ($P_{ss'}(a) \ll 1$) may be poorly resolved relatively.

Theorem 2 (myopic agents are exempt). Even an optimal agent on *every* one-step goal need not determine any nontrivial approximation to P : for the paper’s myopic goal class the only bound on P derivable from π^* is the trivial $\varepsilon = 1$, and this is tight: competence on one-step problems reveals only the comparisons needed to choose actions, not the numerical transition probabilities.

So goal-directedness alone forces nothing; it is *depth*, the need to chain stages, that forces a model. Two qualifications belong right here. First, Theorem 1 establishes *recoverability*: an approximate transition model can be extracted from the policy by an external observer. It does not establish that the agent explicitly represents $\hat{P}$ or computes with it. Second, the converse is the familiar direction—given P , or a good enough $\hat{P}$, goal-optimal policies follow by planning—and it is in this combined sense that the paper says a world model “cannot be avoided”: the information is provably present in the behaviour, whatever the internal mechanism.

Extraction from the policy: the construction behind Theorem 1

As before, the mechanism turns the agent’s *choices* into numbers. Offer a composite goal $\psi_a \vee \psi_b$ whose branches cannot both be pursued: an optimal agent commits to the branch with the larger

maximal success probability, a δ -bounded one to a branch within a factor $(1 - \delta)$ of the best. Each query is thus a slightly noisy comparison of two success probabilities, read from a single action. The reconstruction never runs the environment and never counts realised transitions; it is handed the learned policy and probes it with designed goals. Fix s, s', a and write $p := P_{ss'}(a)$. The proof turns p into the parameter of a binomial distribution and lets the agent's branch choice reveal where that distribution is centred. Two quantities are easy to conflate here, and the paper's notation invites it: n , the goal depth in Theorem 1, and m , the number of repeated transition trials inside the proof. The construction below gives $n = 2m + 1$.

One trial. Define three base goals,

$$R := \Diamond[(s, a)], \quad Y := \bigcirc[(s', \cdot)], \quad F := \bigcirc[(\bar{s}', \cdot)],$$

where $(s', \cdot)$ is the pair-set $\{(s', a') : a' \in \mathbf{A}\}$ and $\bar{s}'$ its complement: R says “eventually be in s and take a ”, Y that the very next state is s' , F that it is not. A successful trial is $\langle R, Y \rangle$, a failed one $\langle R, F \rangle$. Two facts make a trial cost exactly p . The return needs no bookkeeping: the cMP is communicating and the paper shows a deterministic policy reaches any state from any start with probability 1. And the outcome test must be a *next*-goal: $\bigcirc$ tests the single transition after (s, a) , so the outcome is committed the moment a is taken in s , with probability exactly p , and successive trials are independent by the Markov property. Written with $\Diamond$ instead (“eventually reach s' ”), a communicating process would let the agent retry until it succeeded, every such goal would have optimal success probability 1, and it would say nothing about p .

Encoding m trials as a binomial. A success/failure history is a string $x \in \{0, 1\}^m$; with $O_1 := Y$, $O_0 := F$, each is a sequential goal

$$\psi_x = \langle R, O_{x_1}, R, O_{x_2}, \dots, R, O_{x_m} \rangle, \quad \text{e.g.} \quad \psi_{101} = \langle R, Y, R, F, R, Y \rangle,$$

with optimal success probability $p^r(1-p)^{m-r}$, $r = \sum_j x_j$. Distinct patterns are mutually exclusive. The grammar allows disjunctions *of* sequences but not disjunctions *inside* them, so “exactly r successes” is an outer disjunction over complete patterns, and tails follow:

$$\Psi_{=r}^{(m)} = \bigvee_{x: \sum_j x_j = r} \psi_x, \quad \Psi_{\leq k}^{(m)} = \bigvee_{r \leq k} \Psi_{=r}^{(m)}, \quad \Psi_{>k}^{(m)} = \bigvee_{r > k} \Psi_{=r}^{(m)},$$

with optimal success probabilities $\binom{m}{r} p^r (1-p)^{m-r}$, $P(K \leq k)$ and $P(K > k)$ for $K \sim \text{Binomial}(m, p)$. The allowed language has encoded the binomial mass function, and the outer disjunction is what packages a whole family of exclusive histories into one goal.

Making the choice observable. Pick two actions $a_L \neq a_R$, prepend the marker $\top[(\cdot, a_L)]$ to every pattern in $\Psi_{\leq k}^{(m)}$ and $\top[(\cdot, a_R)]$ to every pattern in $\Psi_{>k}^{(m)}$, and offer the disjunction $G_k = \mathcal{L}_k \vee \mathcal{R}_k$. The marker costs nothing (R still returns to s with probability 1), but the policy is deterministic and the branches demand different first actions, so $\pi(s_0, G_k) \in \{a_L, a_R\}$ reveals the branch pursued from the very first action, and δ -boundedness gives

$$\text{left chosen} \Rightarrow P(K \leq k) \geq (1 - \delta)P(K > k), \quad \text{right chosen} \Rightarrow P(K > k) \geq (1 - \delta)P(K \leq k).$$

Recovering p . Sweep $k = 0, \dots, m$: $P(K \leq k)$ rises and $P(K > k)$ falls, so the agent switches once, at a threshold k^* . For an optimal agent k^* is a median of $\text{Binomial}(m, p)$, which lies within one of the mean mp , so $\hat{p} \approx k^*/m$ up to $O(1/m)$. Regret blurs the switch over the range of k on which the two tails are within a factor $(1 - \delta)$; binomial concentration (standard deviation $\sqrt{mp(1-p)}$) makes that window $O(\delta\sqrt{m})$ in k , i.e. $O(\delta/\sqrt{m})$ in p .

first action $\rightarrow$ tail comparison $\rightarrow$ switching threshold $k^* \rightarrow$ binomial median $\rightarrow \hat{P}_{ss'}(a)$.

Trials versus depth. Each trial contributes two base goals, R then Y or F , and the proof prepends one marker, so every sequential branch of G_k has depth

$n = 2m + 1, \quad m = \frac{n-1}{2}.$

Trial count is proportional to goal depth, but the two are not the same quantity: a depth- n competence guarantee lets the proof encode $(n - 1)/2$ trials inside one composite goal. The theorem’s constant confirms the bookkeeping: the trial-level bound $\sqrt{p(1 - p)/(m(1 - \delta))}$ becomes, with $m = (n - 1)/2$,

$$\sqrt{\frac{2p(1 - p)}{(n - 1)(1 - \delta)}},$$

exactly Theorem 1; the 2 is the two goals per trial and the $n - 1$ is the marker. This is why depth helps, and it is the role the *continuum* of mixing weights q played before: resolution came from sweeping q continuously; here it comes from repeating a sub-goal m times inside a deeper goal. A myopic agent ($n = 1$, so $m = 0$) has no trial to compare and no resolution, which is Theorem 2. The resulting procedure (the paper’s Algorithm 1) is unsupervised and universal: its only input is π , and it works for every agent and environment meeting the assumptions.

Relation to the previous paper

The two results are halves of one picture. *Robust Agents* concerns *domain* generalisation: adapting to many interventions forces a *causal* model (G, P) . This paper concerns *task* generalisation: achieving many goals in a fixed environment forces a *predictive* model $P_{ss'}(a)$. The authors’ striking corollary: domain generalisation needs *strictly more* knowledge than task generalisation, because a transition function is observational and causal structure—how the environment responds to intervention—is not identifiable from it. A task-general agent need not know the causal graph; a domain-general one must.

Interpretation and consequences

With the recoverability reading fixed above, four consequences follow. *Safety*: a checkable world model can be elicited from any sufficiently capable model-free agent, as many verification and safe-exploration proposals presuppose. *Capability bounds*: building a general agent is at least as hard as learning an accurate world model, so regret-bounded generality is confined to domains with learnable dynamics. *Emergence*: a mechanism for implicit world models in foundation models, since minimising regret across diverse tasks forces one. And a motivation for model-free methods falls away: the model is learned either way.

Two limits for discussion. The result assumes a fully observed, stationary Markov environment, a deterministic policy, and a regret bound uniform over *every* goal up to depth n ; what a partially observed agent must learn about latent state is open; and Theorem 2 draws the boundary sharply—only multi-stage structure forces the model.

Source. J. Richens, D. Abel, A. Bellot and T. Everitt, *General Agents Contain World Models*, ICML 2025, arXiv:2506.01622; Definitions 2–5, Theorems 1–2, the proof of Theorem 1, and Algorithm 1.